

## Supplementary material

### 1. Supplementary S1. Target-label recovery curve: what a small labelled budget buys

#### 1.1. S1.1 Purpose and status

The main text establishes that label-free alignment does not close the residual transfer gap, and that the part of that gap not attributable to contiguous spatial holdout is conditional. CORAL and per-region standardisation recover a minority of the gap at best, and degrade most of the directions that transfer above chance without them (§4.5, Appendix A(n), §5.5). Within the six directions covered here, the direction that transfers above chance raw is Bejís to Muğla, and it is the direction that both interventions help least. The natural constructive question is therefore what a *small number of target labels* buys, since that is the resource label-free machinery cannot substitute for.

The headline of this analysis is in the main text at Appendix A(u). Its ceilings and denominators are computed on the frames as drawn, which §4.4 shows are not comparable across regions; the recovery fractions should be read as within-frame quantities. The full design, the per-budget table and the limits are here, for two reasons. It covers three of the five regions, so it cannot carry a claim at the paper’s stated scope; and it requires labelled target cells, so it is not an alternative transfer protocol but a quantification of the price of the failure the main text documents. It is a supplementary sensitivity result, not a proposed method.

#### 1.2. S1.2 Design

**Regions and directions.** Three experiments are used in all six ordered directions: Manavgat 2021, Bejís 2022 and Muğla 2021. Evia is excluded by the frozen configuration on two recorded grounds (`evia_2021`: out of scope for this analysis; `evia_2021_extended`: a high-prevalence, different-regime sensitivity control rather than an equal-prevalence primary transfer AOI), and Montiferru does not appear in it. Population, feature sets, forbidden-column set and classifier are the manuscript’s primary choices throughout: natural vegetation (`burnable_tree_shrub_grass`  $\wedge$  `valid_for_modeling`), the ten-feature thermal set as the primary family with the four-feature baseline set as secondary, and the canonical random forest (300 trees, `min_samples_leaf` = 3, `class_weight` = "balanced", `random_state` = 42).

**Labelled budget.** The unit of labelling effort is a 10-cell ( $\approx 5$  km) spatial block, assigned before population filtering, identical to the large-block machinery of §3.7. The configuration records why the canonical 2-cell block is not used here: at  $\approx 1$  km a block holds a median of about four cells, which is neither a plausible unit of survey effort nor separable from the evaluation blocks adjacent to it. Budgets are 0, 1, 2, 4, 8, 16 and 32 blocks. Budget 0 is the raw transfer endpoint, which is the source-only model with no target labels. The ceiling is the target-only model evaluated under the same folds.

**Selection and evaluation.** Blocks are drawn under a fixed tier order (blocks containing both classes, then burned-only, then unburned-only), shuffled within tier by a seed derived as `blake2b(schema|source|target|outer_fold|repeat)`. The seed is independent of the budget, of the model family and of any result, so no branch of the selection can react to an outcome. Budgets are nested: the 32-block set contains the 16-block set. Evaluation is 5-fold `StratifiedGroupKFold` grouped on the target’s large blocks, in strict mode. Each budget is repeated 10 times with different block draws (the raw and ceiling endpoints once each, being deterministic), for 3,642 unique fits, matching the configuration’s own expected count.

**Uncertainty.** The interval reported is a **selection interval**: the 2.5th and 97.5th percentiles across the 10 block-selection repeats. It expresses sensitivity to *which* blocks were labelled and nothing else. It is not a bootstrap, it is not a confidence interval, and no  $p$ -values are produced; the diagnostic enforces this distinction with a forbidden-terminology check. There is consequently no comparable interval on the raw endpoint, which is a single deterministic fit.

**Recovery fraction.** Defined as  $(\text{few-shot} - \text{raw}) / (\text{ceiling} - \text{raw})$ , neither clipped nor absolute-valued, so a budget that leaves the model worse than raw transfer reports a negative fraction rather than zero.

### 1.3. S1.3 Result

Selection intervals are wide at the smallest budgets and narrow at the largest. For Bejís  $\rightarrow$  Manavgat they run  $[0.392, 0.484]$  at one block against  $[0.733, 0.748]$  at 32. See the caution in S1.4 on why the upper-budget intervals are narrow.

Three observations follow, and only the first is comfortable.

**The largest budget tested recovers most of the gap in three of six directions.** Thirty-two blocks carry on average 2,700 to 3,000 labelled cells, roughly 7 to 20 % of the target’s natural-vegetation population depending on

Table S1: Few-shot recovery of target ROC-AUC, thermal model, natural-vegetation population. Raw = source-only transfer (budget 0); ceiling = target-only model at the same 10-cell blocking. Values are the mean over 10 block-selection repeats. Read from `recovery_curve.csv`.

| Direction        | Raw   | 1 blk | 2 blk | 4 blk | 8 blk | 16 blk | 32 blk | Ceiling | Recovered at 32 |
|------------------|-------|-------|-------|-------|-------|--------|--------|---------|-----------------|
| Manavgat → Bejís | 0.326 | 0.475 | 0.500 | 0.607 | 0.659 | 0.724  | 0.772  | 0.824   | 89 %            |
| Muğla → Bejís    | 0.583 | 0.578 | 0.603 | 0.625 | 0.666 | 0.743  | 0.789  | 0.824   | 85 %            |
| Bejís → Manavgat | 0.444 | 0.454 | 0.435 | 0.504 | 0.575 | 0.637  | 0.743  | 0.797   | 85 %            |
| Muğla → Manavgat | 0.401 | 0.413 | 0.420 | 0.447 | 0.471 | 0.517  | 0.626  | 0.797   | 57 %            |
| Manavgat → Muğla | 0.470 | 0.483 | 0.495 | 0.510 | 0.539 | 0.589  | 0.627  | 0.777   | 51 %            |
| Bejís → Muğla    | 0.618 | 0.576 | 0.577 | 0.578 | 0.598 | 0.637  | 0.666  | 0.777   | 30 %            |

the region. At that budget three directions stand at 85 to 89 % of their target-only ceiling: Manavgat → Bejís and Bejís → Manavgat started below chance, Muğla → Bejís at 0.583, above it. The residual the main text documents is therefore expensive but not structural: it is a shortage of target-conditional information, and target labels supply exactly that. That budget should not be described as modest. Limit 4 below gives the reason: for these AOIs the top budget already contains most of the target’s burned cells.

**The recovery is slow where the transfer is worst.** The two directions into Muğla and Manavgat from Muğla reach only 51 to 57 % at the top budget. These are the directions whose raw transfer sits furthest below the ceiling, and Muğla → Manavgat is still below 0.5 AUC after 8 labelled blocks. A larger concept gap costs more labels, not the same labels.

**Small budgets actively hurt the one direction that already transfers.** Bejís → Muğla is the pair that transfers above chance raw (0.618), and it is the pair few-shot recalibration helps least: the curve is *negative* at 1, 2, 4 and 8 blocks (−0.043 to −0.021 AUC), only overtakes raw at 16, and reaches 30 % at 32, the worst of the six. This is the same asymmetry the main text reports for label-free adaptation (§4.5): where the source model already carries a usable conditional relationship, a small target sample perturbs it before it can replace it. The mechanism differs, since here the target labels are real information rather than a covariate rescaling. The direction of the effect is nevertheless the same, and it is the one direction where the intervention is a liability at every budget a field campaign would plausibly afford.

#### 1.4. S1.4 Methodological limits

These are stated so the analysis is not read as more than it is.

1. **Three regions, six directions.** Evia and Montiferru are absent, so this covers six of the twenty directed pairs in the main analysis and cannot speak to the five-region scope.
2. **It requires labelled target cells.** The budget axis is labels in the target region. Nothing here is a label-free method and nothing here weakens the paper’s negative result about label-free alignment. It prices that result.
3. **No joint analysis with the conditional index.** Whether the labelled budget needed to reach a given recovery fraction is predicted by the conditional sign-agreement index of §3.11 was not computed. With six directions it would in any case be a description rather than a test.
4. **The interval is a selection interval, and it narrows for a reason that is not precision.** Bejís holds only 15 blocks containing both classes and 19 containing any burned cell; Manavgat 26 and 28; Muğla 60 and 70. At 16 and 32 blocks the tiered draw has nearly exhausted the both-class blocks, so repeats select almost the same set. The mean labelled-positive count into Bejís is identically 860.0 at 16 blocks and 880.0 at 32 across both source regions. The narrow upper-budget intervals therefore reflect a saturated selection pool, not a well-estimated quantity, and the top budget is not a small budget for these AOIs: 880 of Bejís’s 1,100 burned cells are inside it.
5. **The ceiling is the 10-cell-block target-only value,** not the  $\approx 1$  km within-region headline reported in the main paper, and is correspondingly lower (0.777 to 0.824 against 0.859 to 0.918). Recovery fractions are only interpretable against this matched-blocking ceiling.
6. **Ceiling reproduction verified for all three targets.** Manavgat, Bejís and Muğla all reproduce the frozen large-block artefacts exactly, at an absolute difference of 0.0 against a  $10^{-9}$  tolerance in both families. An earlier draft of this supplement recorded Muğla as unverified, because at that time no frozen block-10 artefact had been located for it; one is now referenced in the export and the corresponding validator check passes.
7. **Provenance.** The export of record was regenerated on 2026-08-14 at commit 6d7a6a71 under scikit-learn 1.9.0, pandas 3.0.2 and NumPy 2.4.4 — the same scikit-learn version to which every other number in this paper is fixed. Its validator reports **67 PASS, 0 FAIL and 0 SKIPPED**. Two files in the export directory, `repeat_metrics.csv` and `oof_predictions.parquet`, retain their 2026-08-03 timestamps

and were not regenerated; every value quoted in Table S1 was checked against the regenerated export.

### *1.5. S1.5 Figure*

A recovery curve is the natural presentation, and it is prepared from `recovery_curve.csv` (thermal family, `metric = roc_auc`). Budget goes on a log-2 axis and target ROC-AUC on the ordinate. Each direction is one line, the selection interval is drawn as a band, and each direction's ceiling is a horizontal reference. Numbers are as in Table S1.

## **References**